Nagar YuwakShikshanSanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to RashtrasantTukadojiMaharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
NAAC A++
Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu
Department of Computer Science & Engineering (IoT)

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.

Session 2026-2027

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge,Problem analysis, Design/development of solutions,Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World,Ethics, Individual and Collaborative Team work,Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Sheikh Wasimuddin
Name of Student

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Session	**2026-27 (ODD)**	**Course Name**	**Lab : Embeded Systems**
Semester	**7**	**Course Code**	**23IOT1706**
Roll No	**61**	**Name of Student**	**Sheikh Wasimuddin**

Practical Number	**09**
Course Outcome	**CO1: Explain the fundamentals of embedded microcontrollers, development environments, and simulation tools.** **CO2: Develop embedded applications using Arduino/ESP32 and interface sensors, actuators, and peripherals.** **CO3: Implement event-driven and communication-based embedded systems using interrupts, UART, and I2C protocols.** **CO4: Design, simulate, and test embedded system applications for real-world applications.**
Aim	Mini Project: Design and Simulation of an IoT-Enabled Embedded System Using Arduino/ESP32/ARM
Problem Statement	IoT-Based Accident Detection and Emergency Alert System
Theory (100 words)	An **IoT-Based Accident Detection and Emergency Alert System** is an embedded system designed to detect vehicle accidents automatically and provide an emergency alert to the concerned person. The system can be developed using a microcontroller such as **Arduino, ESP32, or ARM** along with sensors like an accelerometer, vibration sensor, or gyroscope. These sensors continuously monitor the movement and condition of the vehicle. When an accident or sudden impact is detected, the microcontroller processes the sensor data and determines whether the event is abnormal. If the detected values cross a predefined threshold, the system activates an emergency response. A **GPS module** can be used to obtain the location of the accident, while an **IoT communication module or ESP32 Wi-Fi connectivity** can transmit the alert to a remote user or cloud platform. The system can also include a **buzzer or LED** to provide a local warning. By combining embedded processing, sensors, communication, and IoT technology, the system enables quick accident detection and emergency notification. The proposed system can help reduce the time required to inform emergency contacts and can be useful for improving vehicle and road

	safety.
Procedure and Execution	The project is implemented and simulated in **Wokwi** using an ESP32, MPU6050 accelerometer, OLED display, LED, buzzer, Wi-Fi, and MQTT communication. The provided Wokwi project uses an acceleration threshold of **20 m/s²** for accident detection. 1. Open the Wokwi in a web browser 2. Check the circuit connections in the Wokwi diagram. The main components used are an **ESP32, MPU6050 accelerometer, 0.96-inch OLED display, LED, and buzzer**. 3. Connect the MPU6050 to the ESP32 through the **I2C interface**, using GPIO 21 as SDA and GPIO 22 as SCL. 4. Connect the buzzer to **GPIO 25** and the LED to **GPIO 27** of the ESP32. 5. Open the Arduino code provided in the Wokwi project and verify the required libraries such as Wire, WiFi, PubSubClient, Adafruit_MPU6050, and Adafruit_SSD1306. 6. Set the accident detection threshold in the program. In this project, an acceleration value greater than **20.0 m/s²** is treated as an accident condition. 7. Start the Wokwi simulation by clicking the **Play/Start Simulation** button. 8. During normal operation, the MPU6050 continuously measures acceleration, while the OLED displays **"System: NORMAL"** and **"Monitoring..."**. 9. Observe the acceleration values in the Serial Monitor. The program calculates the total acceleration from the X, Y, and Z-axis readings. 10. Simulate a sudden movement or impact using the MPU6050 controls in Wokwi so that the acceleration exceeds the defined threshold. 11. When the acceleration crosses the threshold, the ESP32 identifies the condition as an accident and activates the emergency response. 12. The buzzer and LED are switched ON, and the OLED displays an **accident detected warning** along with the acceleration value. 13. The ESP32 connects to the Wokwi Wi-Fi network and sends an emergency message through the MQTT broker using the topic iot/accident/alert. 14. Observe the Serial Monitor to verify messages such as **"ACCIDENT DETECTED"** and **"IoT ALERT SENT"**. 15. After approximately five seconds, the system resets the accident status, switches OFF the buzzer and LED, and returns to the normal monitoring state.

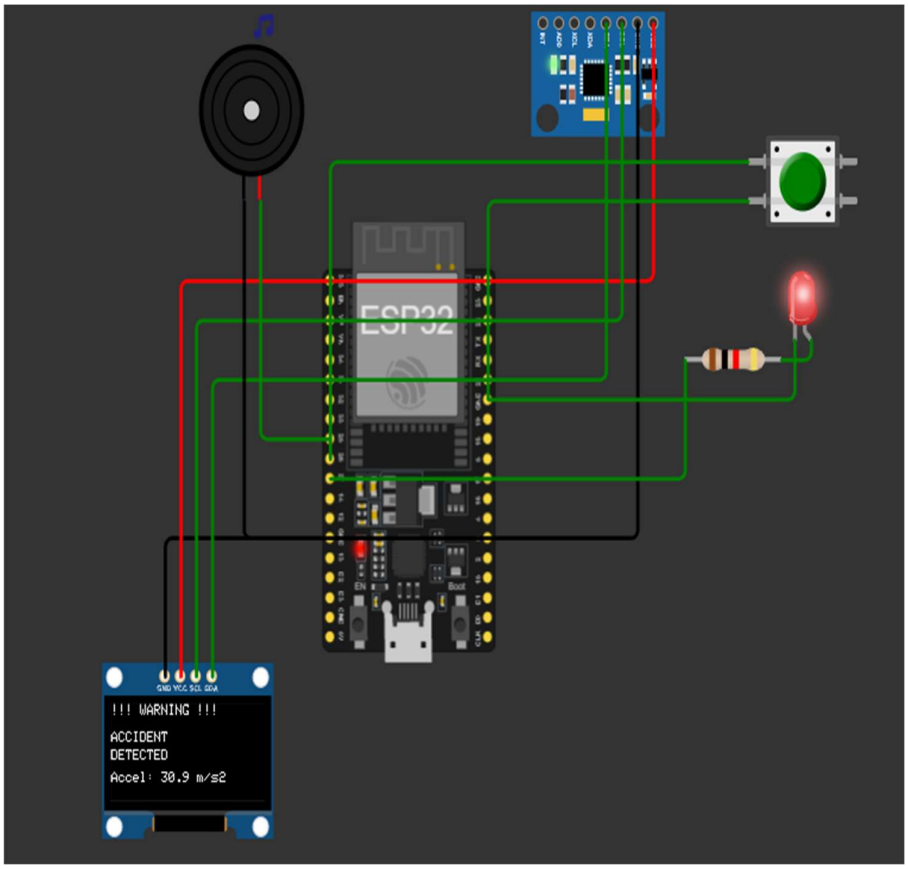
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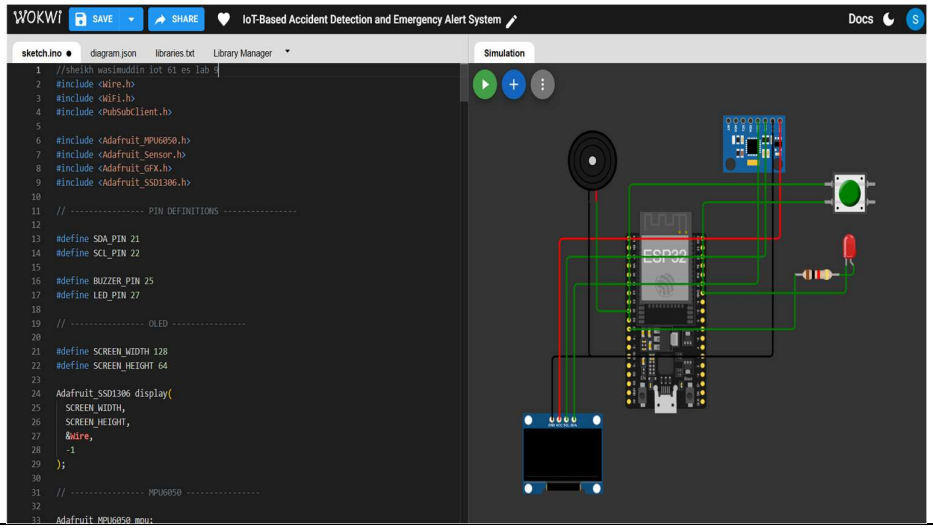
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Stepwise Screenshots with steps:



OFF



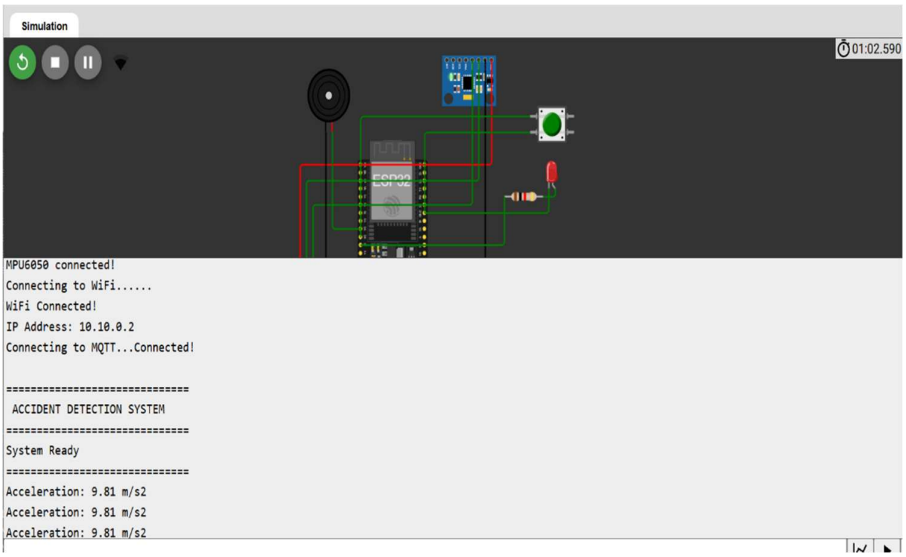
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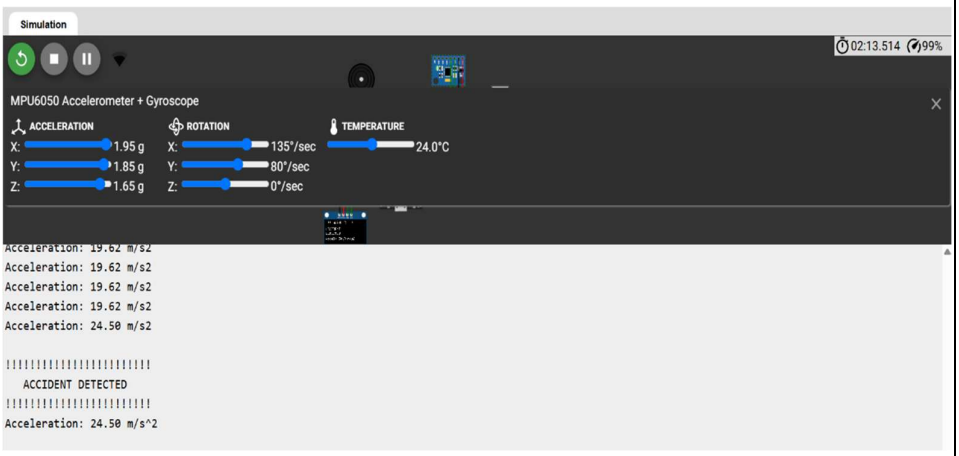
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ON:



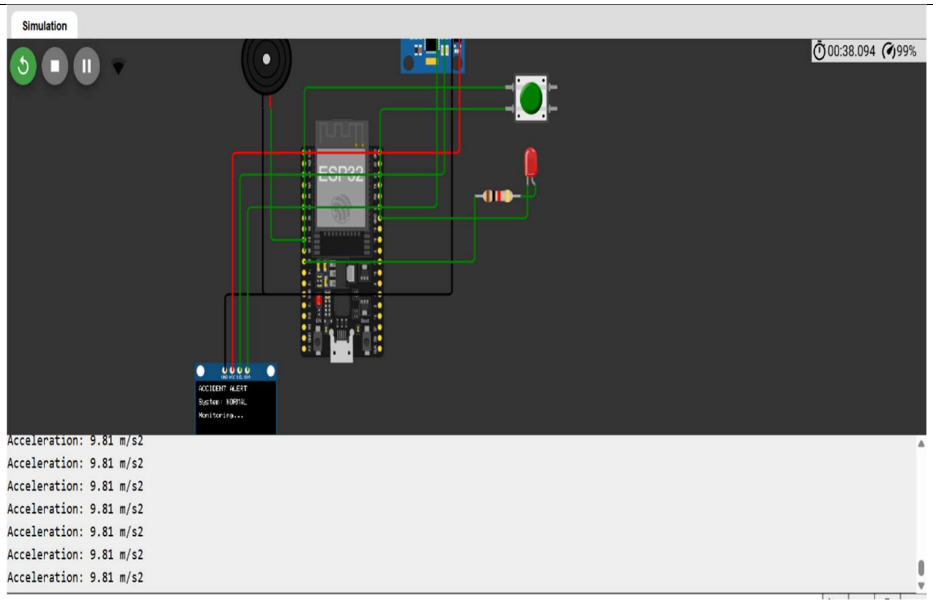


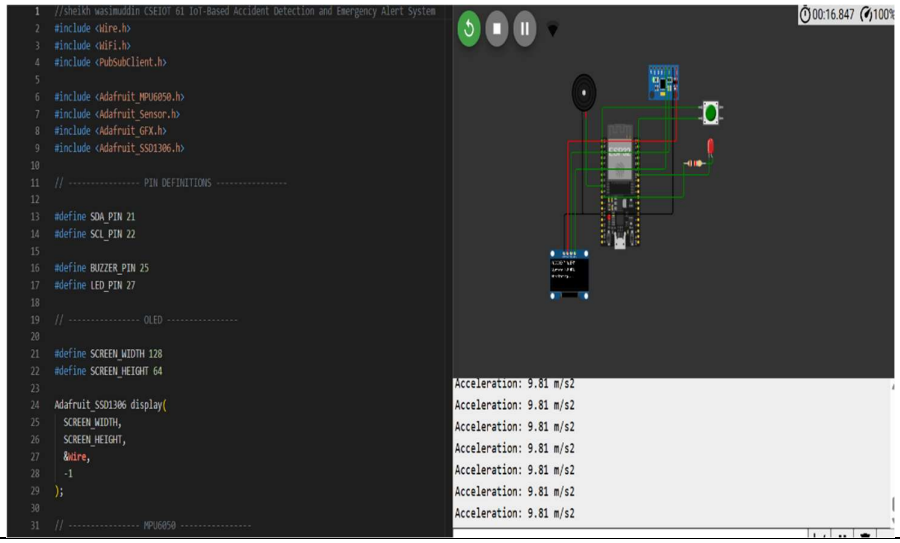
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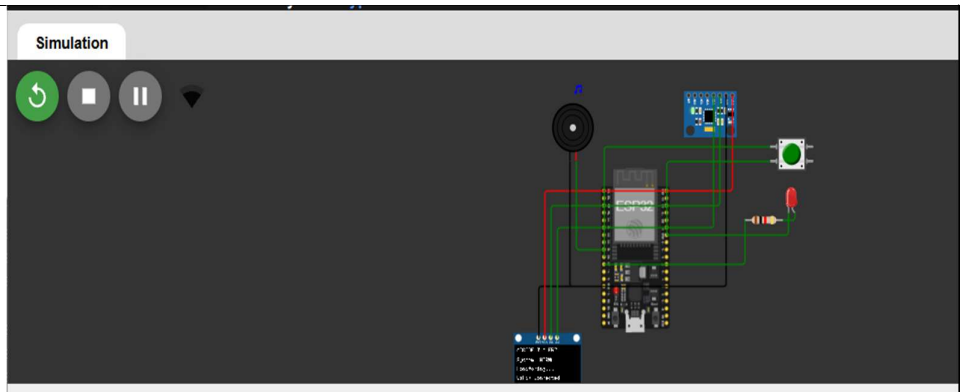


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<table>
<tr><td></td><td> 

IoT ALERT SENT
EMERGENCY ALERT! Accident detected. Acceleration = 24.50 m/s2
Acceleration: 24.50 m/s2
Acceleration: 24.50 m/s2
Acceleration: 28.12 m/s2
Acceleration: 28.12 m/s2
Acceleration: 28.12 m/s2
Acceleration: 28.12 m/s2
Acceleration: 28.12 m/s2
Acceleration: 28.12 m/s2
Acceleration: 30.93 m/s2
Acceleration: 30.93 m/s2
Acceleration: 30.93 m/s2</td></tr>
<tr><td>Conclusion</td><td>The IoT-Based Accident Detection and Emergency Alert System was successfully designed and simulated using ESP32 and Wokwi. The system continuously monitors acceleration using the MPU6050 sensor and detects sudden changes that may indicate an accident. When the acceleration exceeds the predefined threshold, the system activates the LED and buzzer and displays an accident warning on the OLED. The emergency alert is also transmitted using IoT communication through MQTT. Thus, the project demonstrates how embedded systems and IoT technology can be combined to provide automatic accident detection and quick emergency notification.</td></tr>
<tr><td>Date</td><td>24/09/2026</td></tr>
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